Light

GCSE Computer Science — Legal, Ethical & Environmental Issues (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Legal Ethical Environmental

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A company sells software where users must purchase a licence to use it, and the source code is kept secret.
What type of software is this?

- ○ **A.** Proprietary software
- ○ **B.** Open source software
- ○ **C.** Freeware
- ○ **D.** Shareware

Q2 (1 mark)

A person creates a fake website that looks exactly like a bank's login page to steal customers' passwords.
Which law is primarily being broken?

- ○ **A.** The Highway Code
- ○ **B.** The Computer Misuse Act 1990
- ○ **C.** The Food Standards Act
- ○ **D.** The Building Regulations

Q3 (1 mark)

A person downloads cracked versions of expensive software from illegal websites to avoid paying for licences.
What is this practice called?

- ○ **A.** Software updates
- ○ **B.** Cloud computing
- ○ **C.** Software piracy
- ○ **D.** Data backup

Q4 (1 mark)

When visiting a website, a pop-up appears asking permission before storing cookies that track browsing behaviour.
Why is this consent required?

- ○ **A.** To make the website load faster
- ○ **B.** To improve screen resolution
- ○ **C.** To increase storage space

○ **D.** Privacy regulations require informed consent before tracking users' online activities

Q5 (1 mark)

A person asks a company to delete all the personal data it holds about them.
What right under data protection law are they exercising?

- ○ **A.** The right to erasure (right to be forgotten)
- ○ **B.** The right to free internet
- ○ **C.** The right to unlimited storage
- ○ **D.** The right to faster computers

Q6 (1 mark)

Smartphones contain materials like cobalt and tantalum that are sometimes mined in conflict zones under dangerous conditions.
What are these materials called?

- ○ **A.** Recycled materials
- ○ **B.** Conflict minerals
- ○ **C.** Organic compounds
- ○ **D.** Renewable resources

Q7 (1 mark)

The principle that internet service providers should treat all internet traffic equally, without favouring or blocking particular websites or

services.
What is this principle called?

- ○ **A.** Data compression
- ○ **B.** File encryption
- ○ **C.** Net neutrality
- ○ **D.** Cloud storage

Q8 (1 mark)

A student repeatedly sends threatening messages to a classmate through social media, causing distress.
What is this behaviour called?

- ○ **A.** Software debugging
- ○ **B.** Network configuration
- ○ **C.** System administration
- ○ **D.** Cyberbullying

Q9 (1 mark)

UK regulations require businesses to properly dispose of electronic waste and offer collection when selling new equipment.
What are these regulations called?

- ○ **A.** The WEEE Regulations (Waste Electrical and Electronic Equipment)
- ○ **B.** The Highway Code
- ○ **C.** The Building Regulations

- ○ **D.** The Food Safety Act

Q10 (1 mark)

Software that is free to use but does not provide access to its source code is called:

- ○ **A.** Open source software
- ○ **B.** Freeware
- ○ **C.** Shareware
- ○ **D.** Public domain software

Q11 (1 mark)

Most websites and software require users to agree to lengthy terms and conditions before use.
What ethical concern does this raise?

- ○ **A.** The documents load too slowly
- ○ **B.** The font size is too large
- ○ **C.** Users rarely read them, so consent may not be truly informed — they may unknowingly agree to data collection or rights they would reject if understood
- ○ **D.** The documents use too many colours

Q12 (1 mark)

Cryptocurrency mining requires powerful computers running continuously, consuming large amounts of electricity.

What environmental concern does this raise?

- ○ **A.** Cryptocurrency makes computers too colourful
- ○ **B.** Mining equipment is too quiet
- ○ **C.** Computers become too small
- ○ **D.** High energy consumption contributes to carbon emissions, especially when powered by fossil fuels

Q13 (2 marks)

A person submits a Subject Access Request (SAR) to a company, asking for a copy of all personal data the company holds about them.
What must the company do?

- ○ **A.** The company must provide the requested data free of charge within one month, though extensions are possible for complex requests; they must also explain how the data is used and who it is shared with
- ○ **B.** The company can ignore the request entirely
- ○ **C.** The company must delete all the person's data immediately
- ○ **D.** The company must pay the person for their data

Q14 (2 marks)

A disgruntled employee, before leaving their job, deliberately deletes important company files from the server.
Which section of the Computer Misuse Act 1990 applies?

- ○ **A.** No law has been broken because they had legitimate access
- ○ **B.** Section 3 — unauthorised acts with intent to impair the operation of a computer, which covers deliberately damaging or

destroying data

○ **C.** The deletion is only a civil matter, not criminal

○ **D.** The Copyright Act applies instead

Q15 (2 marks)

AI software can now create realistic videos showing people saying or doing things they never actually did.
What ethical and legal concerns do these 'deepfakes' raise?

○ **A.** Deepfakes make videos load faster

○ **B.** Deepfakes improve video resolution

○ **C.** Deepfakes can spread misinformation, damage reputations, enable fraud, create non-consensual intimate imagery, and undermine trust in authentic video evidence

○ **D.** Deepfakes only affect professional actors

Q16 (2 marks)

A movement advocates for laws requiring manufacturers to make devices repairable by providing spare parts, repair manuals, and avoiding deliberate design obstacles.
What is this movement called, and why does it matter environmentally?

○ **A.** The planned obsolescence movement; it encourages faster replacement

○ **B.** The disposable technology movement; it promotes single-use devices

○ **C.** The sealed design movement; it advocates for non-openable cases

○ **D.** The right to repair movement; it extends device lifespan, reduces e-waste, and decreases the environmental impact of manufacturing new replacements

Q17 (2 marks)

A website makes the 'Subscribe to marketing emails' checkbox pre-ticked and hard to find, while the 'No thanks' option is greyed out and barely visible.
What unethical design practice is this?

○ **A.** Dark patterns — deliberately deceptive design that manipulates users into actions they would not otherwise choose, undermining genuine consent

○ **B.** Responsive design — adapting to different screen sizes

○ **C.** Accessible design — supporting users with disabilities

○ **D.** Minimalist design — removing unnecessary elements

Q18 (2 marks)

A software developer wants to protect their new invention — a unique algorithm that dramatically speeds up video compression.
What forms of intellectual property protection might apply?

○ **A.** Only trademarks can protect software

○ **B.** Patents can protect novel inventions (including some software-implemented methods), while copyright automatically protects the specific code expression; the algorithm's implementation is copyrighted, and the inventive method may be patentable

○ **C.** No intellectual property protection exists for software

○ **D.** Only physical inventions can be protected

Q19 (2 marks)

A company moves its IT systems from local servers to cloud computing, claiming this is more environmentally friendly.
Is this claim accurate?

○ **A.** Cloud computing has no environmental impact at all

○ **B.** Cloud computing always has worse environmental impact than local servers

○ **C.** Cloud computing can be more efficient due to data centre optimisation and economies of scale, but environmental benefits depend on the cloud provider's energy sources and the company's actual usage patterns

○ **D.** Environmental impact only depends on the colour of the servers

Q20 (2 marks)

Citizens argue they should be able to understand how algorithms used by government services make decisions that affect their lives, such as benefit eligibility or risk assessments.
What is this demand called?

○ **A.** Algorithmic secrecy — keeping all algorithms hidden

○ **B.** Algorithmic complexity — making systems more complicated

○ **C.** Algorithmic speed — making systems faster

○ **D.** Algorithmic transparency — the ability to understand and scrutinise automated decision-making that affects people's lives

Q21 (2 marks)

A job applicant is rejected after the employer searches online and finds embarrassing social media posts from years ago.
What does this illustrate about digital footprints?

- ○ **A.** Online information persists indefinitely, can be discovered in unexpected contexts, and may affect opportunities years later — individuals have limited control over how past posts are perceived
- ○ **B.** Digital footprints only exist for 24 hours
- ○ **C.** Employers cannot legally access any social media
- ○ **D.** Online posts automatically delete after one year

Q22 (2 marks)

A consumer debates whether streaming films has less environmental impact than buying physical DVDs.
What factors affect this comparison?

- ○ **A.** Only the colour of DVD cases matters
- ○ **B.** Streaming has zero environmental impact
- ○ **C.** Comparison depends on: DVD manufacturing and shipping impacts vs streaming's data centre energy, network transmission, and device usage; watching once favours streaming while repeated viewing may favour physical media; energy sources matter significantly
- ○ **D.** Physical media always has zero environmental impact

Q23 (2 marks)

An employee discovers their company is illegally dumping toxic waste and reports this to regulators.

How do legal protections for whistleblowing interact with data protection and confidentiality obligations?

- ○ **A.** Whistleblowers are never protected and always face prosecution
- ○ **B.** Whistleblowing protections override normal confidentiality in the public interest — the Public Interest Disclosure Act protects employees who report wrongdoing, and data protection allows disclosure to prevent crime
- ○ **C.** All company information must remain confidential regardless of illegality
- ○ **D.** Whistleblowing is only permitted for minor issues

Q24 (3 marks)

A UK resident's personal data is misused by a social media company headquartered in the USA with European operations based in Ireland. How do data protection enforcement challenges arise in this situation?

- ○ **A.** Data protection only applies within single countries
- ○ **B.** All complaints must be resolved within 24 hours
- ○ **C.** No enforcement is possible for international companies
- ○ **D.** Jurisdictional complexity: UK regulator (ICO) has limited direct authority over US companies; 'one-stop-shop' mechanisms may route complaints through Irish regulators; companies may exploit jurisdictional gaps; enforcement requires international cooperation that may be slow or ineffective

Q25 (3 marks)

AI systems can now generate text, images, and music that appear original but are trained on existing copyrighted works.
What legal and ethical questions does AI-generated content raise?

- ○ **A.** AI content has no legal implications because computers cannot create
- ○ **B.** Training on copyrighted works without permission may infringe copyright; ownership of AI outputs is unclear (creator? user? AI company?); AI-generated work may not qualify for copyright protection itself; artists whose styles are replicated without consent raise moral rights concerns; existing copyright law struggles with these novel situations
- ○ **C.** All AI-generated content automatically belongs to the public domain
- ○ **D.** Copyright law already fully addresses all AI-related issues

Q26 (3 marks)

Police use facial recognition technology to scan crowds at public events, matching faces against databases of wanted individuals.
What ethical considerations should govern use of this technology?

- ○ **A.** Technology efficiency is the only relevant consideration
- ○ **B.** Accuracy (false positives disproportionately affect ethnic minorities), proportionality (mass surveillance vs targeted use), consent (public cannot opt out), bias (training data may embed discrimination), chilling effects (people may avoid lawful gatherings if watched), oversight (who authorises use, reviews matches, handles errors), and fundamental questions about surveillance in democratic society
- ○ **C.** No ethical considerations apply to law enforcement technology
- ○ **D.** Public spaces should have no expectation of privacy whatsoever

Q27 (3 marks)

Platform companies classify delivery drivers as 'self-employed independent contractors' rather than employees, meaning they lack

employment protections like minimum wage, sick pay, and holiday entitlement.
What ethical issues does this classification raise?

- ○ **A.** Classification is purely a technical legal matter with no ethical dimensions
- ○ **B.** All workers should always be classified as self-employed for flexibility
- ○ **C.** Worker classification affects fundamental fairness: if platforms control work through algorithms (accepting jobs, ratings, deactivation), treating workers as independent may misrepresent the actual relationship; workers bear risks while platforms retain control; legal classifications may not reflect working realities; vulnerable workers may lack bargaining power to secure fair treatment
- ○ **D.** Employment law has no application to technology companies

Q28 (3 marks)

Social media platforms must decide whether to remove content that some consider harmful but others defend as free speech, such as controversial political opinions or disputed health claims.
How should platforms balance these competing concerns?

- ○ **A.** Platforms should never remove any content regardless of harm
- ○ **B.** Platforms should remove all controversial content to avoid complaints
- ○ **C.** All content decisions should be made by government without platform involvement
- ○ **D.** Balancing requires: transparent policies developed with diverse input; consistent enforcement (not arbitrary); distinguishing illegal content, provable falsehood, and contested opinion; appeals processes for errors; recognising both over-moderation (silencing

legitimate speech) and under-moderation (enabling harm) cause damage; accountability for decisions affecting public discourse

Q29 (3 marks)

Technology companies employ psychologists and use techniques like variable rewards, infinite scroll, and notification timing to maximise user engagement, which critics argue deliberately creates addictive products. What ethical responsibilities should technology companies have regarding addictive design?

- ○ **A.** Engagement maximisation techniques should be designed with awareness of psychological effects, particularly on vulnerable users (children, those prone to addiction); companies should provide tools for users to manage usage; design should support user wellbeing, not exploit psychological vulnerabilities; profit from engagement should not override user welfare; transparency about persuasive techniques enables informed choice
- ○ **B.** Companies have no responsibility for how users interact with products
- ○ **C.** Addictive design is always beneficial because it proves users enjoy products
- ○ **D.** Only users are responsible for their technology usage

Q30 (3 marks)

A 'circular economy' approach to electronics would involve designing products for longevity, repairability, refurbishment, and eventual material recovery, rather than the current 'take-make-dispose' linear model. What changes would be needed to implement circular economy principles in the technology industry?

- ○ **A.** Current practices are optimal and require no change
- ○ **B.** Design changes (modular repairable devices), business model shifts (product-as-service, leasing), extended producer responsibility

(manufacturers manage end-of-life), consumer behaviour changes (valuing longevity over novelty), regulatory requirements (durability standards, repairability indices), and infrastructure development (collection, refurbishment, recycling) — all requiring coordination across the value chain

○ **C.** Only consumers need to change; manufacturers are not responsible

○ **D.** Electronics cannot be recycled under any circumstances

Total: **55** marks • Time: **30 min**

↑ Back to top

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